

Deriving the Quadratic equation

Emil Kerimov

September 28, 2026

1 The Quadratic Equation

Given a quadratic polynomial $ax^2 + bx + c = 0$ The two solutions of x that satisfy this equation are given by the quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

2 Algebraic Proof

The most straight-forward proof is given by a technique referred to as 'Completing the Square'.

$$ax^2 + bx + c = 0$$

Dividing both sides by a

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Adding and subtracting $\frac{b^2}{(2a)^2}$

$$x^2 + \frac{b}{a}x + \frac{b^2}{(2a)^2} + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Recognizing the 'square'

$$\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Rearranging and solving

$$\begin{aligned} \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2} \\ x + \frac{b}{2a} &= \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\ x &= -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$